Category
**Pricing and Valuation of Options**

Related Questions:
**Practice questions related to this topic**

# FACTORS AFFECTING OPTION VALUE

## Learning Outcome

- identify the factors that determine the value of an option and describe how each factor affects the value of an option

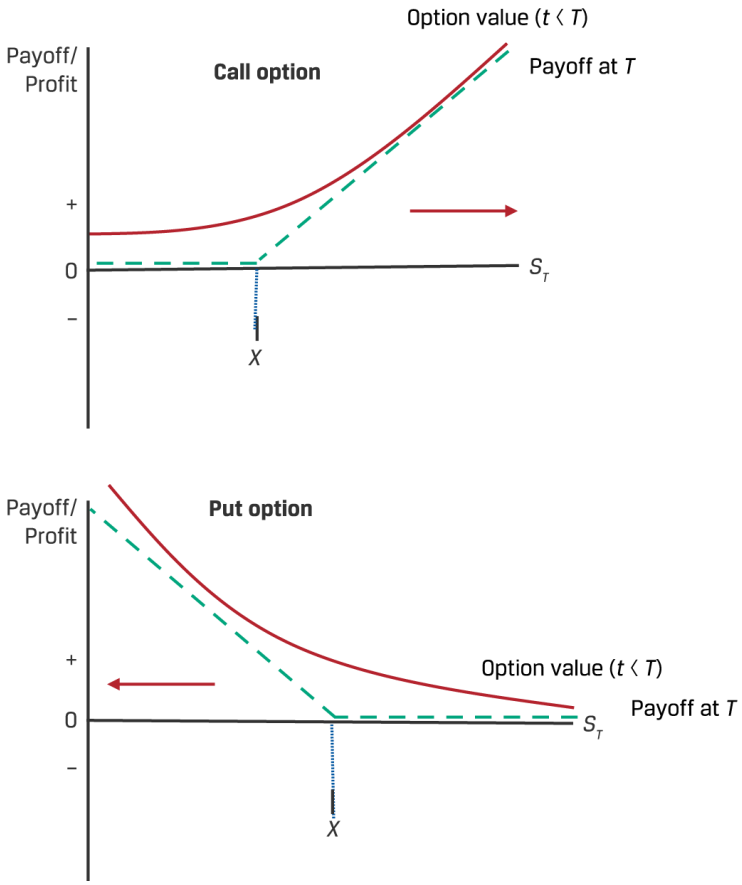
We now turn our attention to identifying and describing factors that affect an option's value. Several of these factors are common to both forward commitments and options, and a number are unique to contingent claims. Factors that determine the value of an option include the value of the underlying, the exercise price, the time to maturity, the risk-free rate, the volatility of the underlying price, and any income or cost associated with owning the underlying asset.

## Value of the Underlying

Changes in the value of the underlying will have the same directional effect on the *right* to transact the underlying under an option contract as on the *obligation* to transact under a forward commitment. For example, a call option and a long forward position will both *appreciate* if the spot price of the underlying *rises*, while a put option and a short forward position will both *appreciate* if the spot price of the underlying *falls*, as shown in Exhibit 6.

**Exhibit 6:**

**Call and Put Option Value versus Underlying Value**




▶ Expand / collapse Extended Description

price, while for an option it is a *non-linear* relationship that depends on the likelihood that an option buyer will exercise in the future. The more likely it is that the option buyer will exercise (e.g., when the option is in the money), the more sensitive the option's value to the underlying price. This likelihood depends on the relationship between the value of the underlying and the exercise price, a feature unique to option contracts that we turn to next.



## Exercise Price

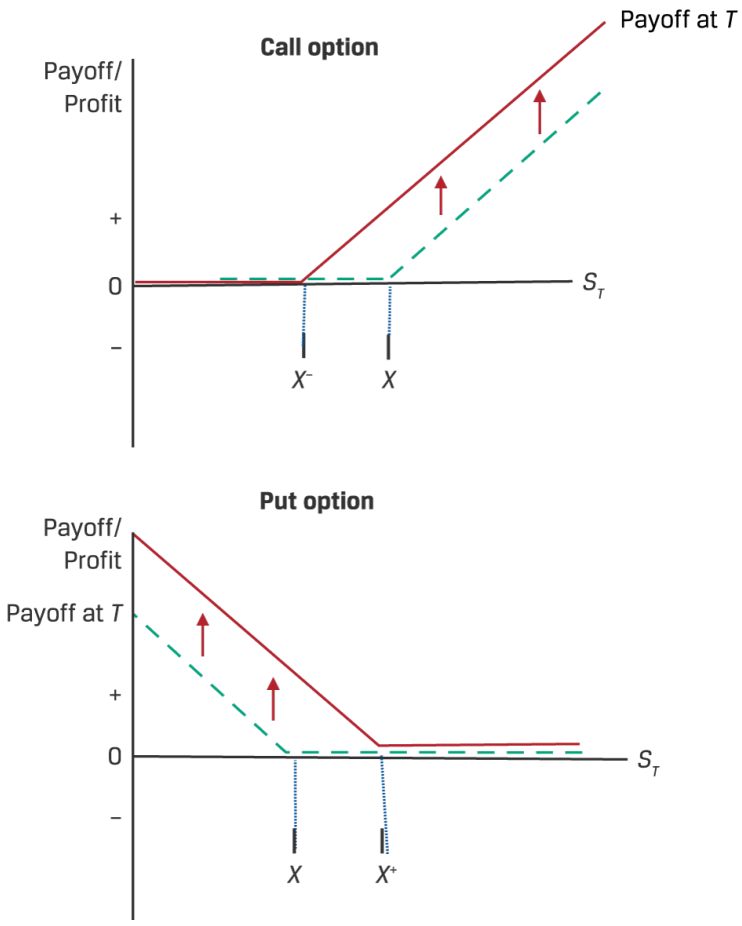
The exercise price is the threshold that determines whether an option buyer chooses to transact at contract maturity. For a call option representing the right to buy the underlying, the exercise price represents a *lower* bound on the option's exercise value at maturity, leading to a *higher* option value for a *lower* exercise price. As call options settle based on Max (0, $S_T - X$) at expiry, a lower $X$ will increase both the likelihood of exercise and the settlement value if exercised.

The exercise price of a put option, in contrast, is an *upper* bound on its exercise value at maturity. A *higher* exercise price at which a put option buyer has the right to sell the underlying will therefore *increase* the value of the option. Exhibit 7 shows these differing effects of exercise price changes for both call and put options.



**Exhibit 7:**

**Call and Put Option Value versus Exercise Price**




▶ Expand / collapse Extended Description

## Time to Expiration

Recall from a prior lesson that the time value, or difference between an option's price and its exercise value, represents the likelihood that favorable changes to the underlying price will increase the profitability of exercise. Unlike a long forward position where the buyer is

For a call option, a longer time to expiration will increase the option's value in *all* cases. The price appreciation potential of an underlying is essentially unlimited and increases over longer periods, while the downside is limited to the loss of the premium.

A put option representing the right to sell the underlying also usually benefits from the passage of time. In the case of a put option, a longer time to expiration offers greater potential for price *depreciation* below the exercise price, while the loss is limited to the premium if the underlying price rises. However, as put option buyers await the *sale* of the underlying in order to receive ($X - S_T$) upon exercise, a longer time to expiry *reduces* the present value of the payoff. While less common, in some cases a longer time to expiration will *lower* a put's value, especially for deep-in-the-money puts with a longer time to expiration and a higher risk-free rate of interest.

## Risk-Free Interest Rate

An earlier lesson showed the risk-free rate to be the opportunity cost of holding an asset, which extends to the no-arbitrage valuation of derivatives. For example, an option's exercise value was shown to be equal to the difference between the spot price and the *present value* of the exercise price:

Call Option Exercise Value: Max (0, ($S_t$ − PV($X$)))

Put Option Exercise Value: Max (0, (PV($X$) − $S_t$))

A *higher* risk-free rate therefore *lowers* the present value of the exercise price *provided* an option is in the money. A higher risk-free rate will *increase* the exercise value of a call option and *decrease* the exercise value of a put option. Note that the risk-free rate does not directly affect the time value of an option.

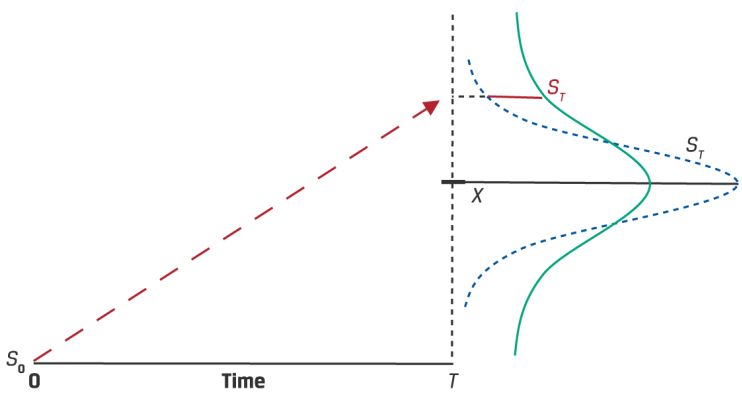
## Volatility of the Underlying

Volatility is a measure of the expected dispersion of an underlying asset's future price movements. Higher price volatility of the underlying increases the likelihood of a higher positive exercise value without affecting the downside case in which the option expires unexercised, as shown in Exhibit 8. Note that this effect will be the same for both call options and put options.

For example, as volatility *rises*, a wider range of possible underlying prices *increases* an option's time value and the likelihood that it will end up in the money. *Lower* volatility will *reduce* the time value of both put and call options.

**Exhibit 8:**

**Volatility and Option Value**




▶ Expand / collapse Extended Description

## Income or Cost Related to Owning Underlying Asset

Rate Your Confidence

Category
**Pricing and Valuation of Options**

Related Questions:
**Practice questions related to this topic**

asset but not to the owner of a derivative. This effect holds for both forward commitments and contingent claims. Income or other benefits of ownership *decrease* the value of a call and *increase* the value of a put. Carry costs (such as storage and insurance for commodities) have the opposite effect, *increasing* the value of a call option and *decreasing* the value of a put. A summary of the factors that affect an option's value is provided in Exhibit 9, with the sign on the option value referring to the impact (positive or negative) of an *increase* in the factor.



### Exhibit 9:

### Factors That Affect the Value of an Option

| Factor | Call Value | Put Value |
|---|---|---|
| Value of the underlying | + | − |
| Exercise price | − | + |
| Time to expiration | + | +/− |
| Risk-free interest rate | + | − |
| Volatility of the underlying | + | + |
| Income/cost related to owning the underlying | −/+ | +/− |



## Question Set: Factors That Affect Option Value

1. Determine the correct answers to complete the following sentence: For a call option representing the right to buy the underlying, the exercise price represents a _____ bound on an option's exercise value at maturity, leading to a _____ option value for a lower exercise price.

Hide Solution

**Solution:**

For a call option representing the right to buy the underlying, the exercise price represents a *lower* bound on an option's exercise value at maturity, leading to a *higher* option value for a lower exercise price.

2. Match the following changes in factors affecting option value (holding other factors constant) with their corresponding option value change:

| | |
|---|---|
| 1. A lower exercise price ($X$) | A. Increases the value of both a call option and a put option |
| 2. A lower underlying price ($S_T$) | B. Increases the value of a call option |
| 3. A rise in the volatility of the underlying price | C. Increases the value of a put option |

Hide Solution

**Solution:**

1. B is correct. A lower exercise price increases the value of a call option, since for a given underlying price at maturity of $S_T$, the call option settlement value of Max (0, $S_T - X$) will increase for a lower $X$.
2. C is correct. A lower underlying price ($S_T$) will increase the value of a put option. Since a put option is the right to sell an underlying at the exercise

value of both a call option and a put option. Greater price variability of the underlying will increase the probability of a higher positive exercise value for a call or a put option without affecting the downside case where the option expires unexercised.

3. Determine the correct answers to complete the following sentences: A key distinction between forward commitments and contingent claims is the _____ of a derivative's price change for a given change in the underlying value. For a forward commitment, the derivative value is a _____ function of the underlying price, while for an option it is a _____ relationship.

Hide Solution

**Solution:**

A key distinction between forward commitments and contingent claims is the *magnitude* of a derivative's price change for a given change in the underlying value. For a forward commitment, the derivative value is a *linear* function of the underlying price, while for an option it is a *non-linear* relationship.

4. Explain the effect of a *decrease* in the risk-free rate of interest on the value of put and call options and justify your answer.

Hide Solution

**Solution:**

A *decrease* in the risk-free rate *increases* the present value of an option's exercise *price* (PV($X$)). The exercise value for an *in-the-money call* option will fall, since ($S_t$ − PV($X$)) is lower for a higher PV($X$), while the exercise value for an *in-the-money put* option will rise, since (PV($X$) − $S_t$) rises for a higher PV($X$). Note that the time value of an option is not directly affected by the risk-free rate.

Rate Your Confidence

Category

**Pricing and Valuation of Options**

Related Questions:

**Practice questions related to this topic**

**OPTIONAL UNTESTED SEGMENT**

## UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

### {OVME} OPTION VALUATION EQUITY

The Bloomberg OVME function allows you to value options.

> "OVME <GO> allows you to price over-the-counter derivative contracts for equity underlyings, so you can accurately quantify your market exposures for options on single-name stocks, equity indices, and funds. Using an assortment of pre-trade analytical tools, you can examine curves and cashflows and perform risk and scenario analyses for listed or OTC plain vanilla options, bond futures, listed or OTC warrants, short-term interest rate futures (STIRs), and exotic options such as variance swaps. OVME uses interpolated market-implied volatilities when the underlying has an active market of listed options, allowing you to replicate the market conditions at the time of the trade. OVME also provides trading tools that allow you to manage and execute the trades you work up in the function." -Bloomberg

To explore this function, type OVME into the Bloomberg command line and press Enter or <GO>. Type a security—such as China Life Insurance Company Ltd. (or 2628 HK Equity)—into the top left "Underlying" amber field, and press Enter or <GO>. The amber filters are

a call or a put, and what style of option it is. See the Delta, Gamma, Vega, Theta, and Rho values for an option. See the implied volatility used to price the option in the "Vol" field.

© 2023 Bloomberg L.P.

Please provide feedback on this Bloomberg feature.

Rate Your Confidence

Category

**Pricing and Valuation of Options**

Related Questions:

**Practice questions related to this topic**

Discuss

Filter

Discussion

What do you want to discuss... (required)

No messages to display

© 2025 CFA Institute. All Rights Reserved.

Rate Your Confidence

Category

**Pricing and Valuation of Options**

Related Questions:

**Practice questions related to this topic**

Rate Your Confidence

Category

**Pricing and Valuation of Options**

Related Questions:

**Practice questions related to this topic**

Rate Your Confidence

Category

**Pricing and Valuation of Options**

Related Questions:

**Practice questions related to this topic**

Rate Your Confidence

Category

**Pricing and Valuation of Options**

Related Questions:

**Practice questions related to this topic**

Rate Your Confidence

Category

**Pricing and Valuation of Options**

Related Questions:

**Practice questions related to this topic**